3) $(4x-3y)dx+(2y-3x)dy=0$

$y=vx \Rightarrow u=\frac{y}{x}$

$dy=vdx+xdu$

$4xdx-3ydx+2ydy-3xdy=0$

$4xdx-3vxdx+2vx(vdx+xdu)-3x(vdx+xdu)=0$

$4xdx-3vxdx+2u^2xdx+2ux^2du-3xvdx-3x^2du=0$

$4xdx-6vxdx+2u^2xdx=3x^2du-2ux^2du$

$4dx-6vdx+2u^2dx=3xdu-2uxdu$

$(4-6u+2u^2)dx \quad =x(3du-2udu)$

$\frac{dx}{x}=\frac{3du-2udu}{4-6u+2u^2}$

$\int\frac{dx}{x}=\int\frac{(3-2u)du}{2u^2-6u+4}$

$\frac{1}{2}\int\frac{3-2udx}{u^2-3u+2} \quad t=u^2-3u+2$

$dt=(2u-3)dx$

$-\frac{1}{2}\int\frac{\cancel{2u-3}}{t}\frac{dt}{\cancel{2u-3}}$

$-\frac{1}{2}\ln|2u-3|+c$

$\ln x+\frac{1}{2}\ln|2u-3|=\ln k$

$2\ln x+\ln|2u-3|=\ln k$

$\ln x^2(2u-3)=\ln k$

$x^2(2u-3)=k$

$x^2[2(\frac{y}{x}-3)]=k$

$2xy-3x^2=k$

$y'=\frac{y+x}{y-x}$ $\quad y=vx$

$dy=vdx+xdu$

$\frac{dy}{dx}=\frac{y+x}{y-x}$

$(y-x)dy=(y+x)dx$

$ydy-xdy=ydx+xdx$

$vx(vdx+xdu)-x(vdx+xdu)=vxdx+xdx$

$v^2xdx+vx^2du-vxdx+x^2du=vxdx+xdx$

$u^2xdx-vxdx-vxdx-xdx=-vx^2du-x^2du$

$u^2xdx-2vxdx-xdx=-vx^2du-x^2du$

$u^2dx-2udx-dx=-vxdu-xdu$

$(u^2-2u-1)dx=x(-vdu-du)$

$\frac{dx}{x}=\frac{-vdu-du}{u^2-2v-1}$

$-\int\frac{v+1dv}{v^2-2v-1}$ $\quad t=v^2-2u-1$

$dt=2v-2$

$$\int \frac{dx}{x} - \int \frac{-u\,du - du}{u^2 - 2u - 1} = 0$$

$\ln|x| -$